



NORTH SYDNEY BOYS HIGH SCHOOL

2017 HSC ASSESSMENT TASK 3 (TRIAL HSC)

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- For Section I, shade the correct box on the sheet provided
- For Section II, write in the booklet provided
- Each new question is to be started on a **new page**.
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Zuber
- ☐ Mr Ireland
- ☐ Mr Berry
- ☐ Ms Ziazaris
- ☐ Mr Hwang

Student Number

(To be used by the exam markers only.)

Question No	1-10	11	12	13	14	15	16	Total	Total
Mark	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{100}$	$\overline{100}$

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10

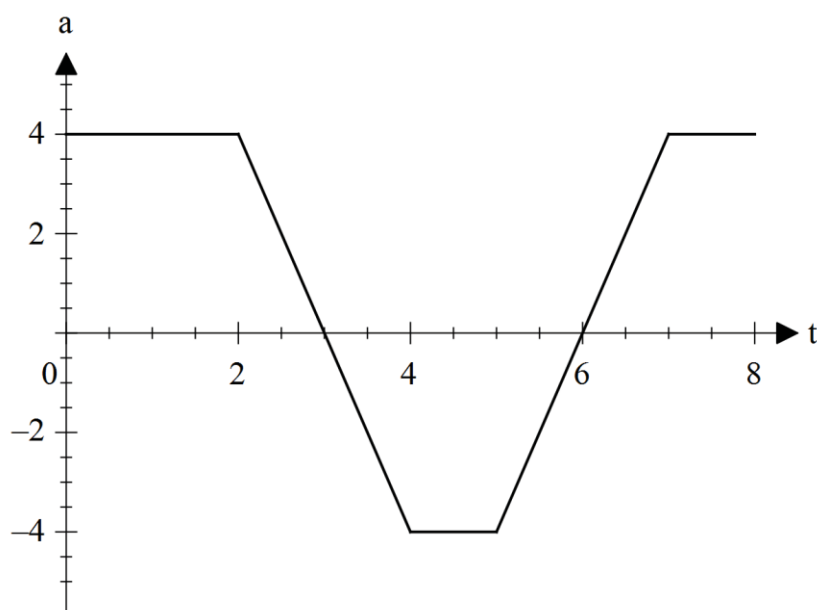
- 1 $x + 2y - (x - 2z) - (2y + x - (2z - x))$ simplifies to
- (A) $4z + 2x$
 - (B) $4z - 2x$
 - (C) $4y + 4z - 2x$
 - (D) $4y - 4z - 2x$
- 2 Which of the following **best** describe the locus of a point which moves so as to always be equidistant from two fixed points?
- (A) a circle
 - (B) a parabola
 - (C) a straight line
 - (D) an exponential
- 3 $\log_3 15 + \log_3 18 - \log_3 10$ evaluates to:
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0

- 4 Find the limiting sum of the following geometric series:

$$-1 + \frac{2}{3} - \frac{4}{9} + \frac{8}{27} + \dots$$

- (A) -3
- (B) 3
- (C) $\frac{3}{5}$
- (D) $-\frac{3}{5}$
- 5 Which of the following most accurately describe the parabola $y = -2x^2 + 9x - 11$?
- (A) Positive definite
- (B) Negative Definite
- (C) Indefinite
- (D) None of the above
- 6 The centre of the circle whose equation is $x^2 - 6x + y^2 + 4y - 3 = 0$ is
- (A) $(6, -4)$
- (B) $(-3, 2)$
- (C) $(0, 0)$
- (D) $(3, -2)$

- 7 The acceleration – time graph of a particle is shown below:



The time(s) when the particle has an absolute minimum velocity is

- (A) $4 < t < 5$
(B) $t = 0$
(C) $t = 3$
(D) $t = 6$
- 8 What is the value of $\int_{-2}^3 |x - 1| dx$?

- (A) $\frac{11}{2}$
(B) $\frac{5}{2}$
(C) $\frac{13}{2}$
(D) $\frac{17}{2}$

- 9 Which of the following is **not** equivalent to $\int_{-a}^a f(x)dx$ when $f(x)$ is **any** odd function?
- (A) 0
- (B) $3 \int_{-a}^a f(x)dx$
- (C) $\int_{-a}^0 f(x)dx + \left| \int_0^a f(x)dx \right|$
- (D) $\int_{-a}^0 f(x)dx - \int_a^0 f(x)dx$
- 10 How many non-similar triangles have angles whose degree measures are distinct positive integers in arithmetic progression?
- (A) 59
- (B) 60
- (C) 89
- (D) 178

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section.

Answer each question on a NEW page. Extra writing booklets are available.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW page in your writing booklet.

(a) Express $\frac{1}{\sqrt{5} - \sqrt{2}}$ with a rational denominator 2

(b) Solve $|x - 1| < 3$ and graph the solution on a number line. 2

(c) The arc of a circle subtends an angle of 100° at the centre. If the radius of the circle is 12cm, calculate the exact length of the arc. 2

(d) Differentiate the following:

(i) $\tan 2x$ 1

(ii) $x^2 \log_e x$ 2

(e) Find the following integral:

$\int x^3 + \frac{1}{2x} dx$ 2

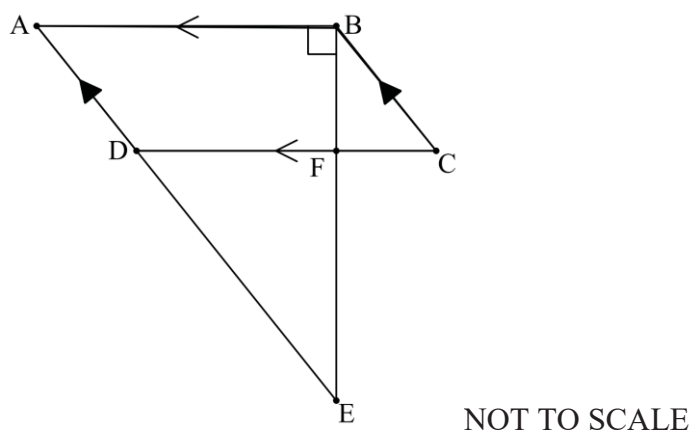
(f) Evaluate the following integral

(i) $\int_0^{\frac{2\pi}{3}} \sin\left(\frac{x}{2}\right) dx$ 2

(ii) $\int_1^2 e^{4x} + e^{-x} dx$ 2

Question 12 (15 marks) Start a NEW page in your writing booklet.

(a)

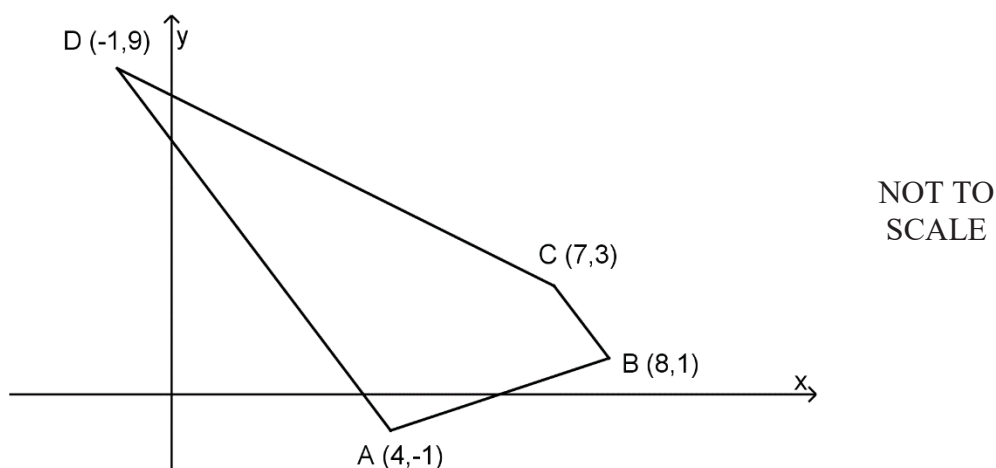


$ABCD$ is a parallelogram where $FB \perp AB$.

(i) Prove that $\triangle CBF \parallel \triangle AEB$ 3

(ii) If $CF = 3\text{cm}$ and $BC = 7\text{cm}$ and $AE = 15\text{ cm}$, find the length of AB 2

(b)



A, B, C and D are the points $(4, -1)$, $(8, 1)$, $(7, 3)$ and $(-1, 9)$ respectively.

(i) Show the equation of AC is $4x - 3y - 19 = 0$ 2

(ii) Show $BC \parallel AD$ 1

(iii) Show $\angle ACD = 90^\circ$ 1

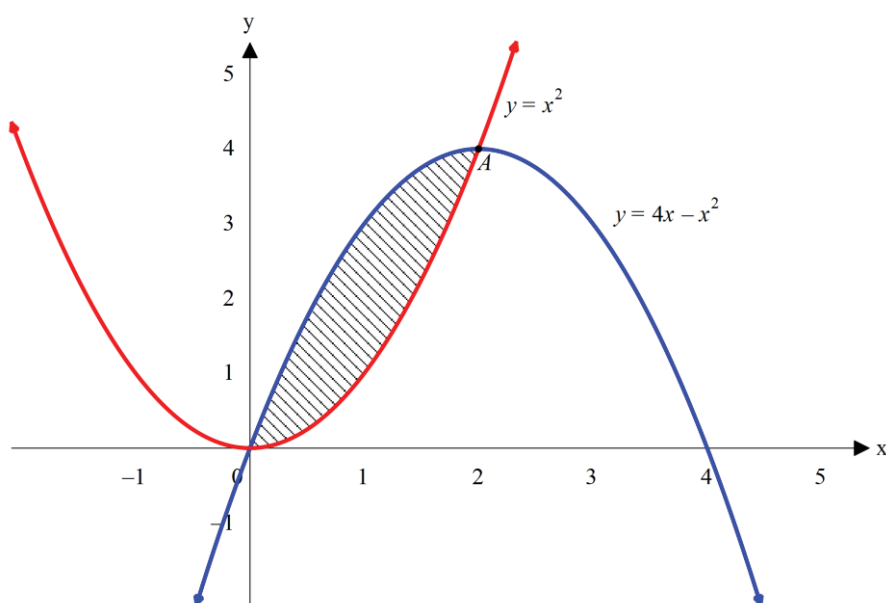
(iv) Show the length of AC is 5 units. 1

(v) Find the perpendicular distance of B from AC 2

(vi) Find the area of the trapezium $ABCD$ 3

Question 13 (15 marks) Start a NEW page in your writing booklet.

- (a) The equation of a parabola is $x^2 = 12y + 6x + 15$. Find
- (i) the coordinates of the vertex 2
 - (ii) the coordinates of the focus 1
 - (iii) the equation of the directrix 1
- (b) For what values of p does the equation $x^2 - px + p - 1 = 0$ have
- (i) equal roots. 2
 - (ii) one of the roots equal to 3 2
- (c) The diagram shows the curves $y = x^2$ and $y = 4x - x^2$, which intersect at the origin and at the point A.



- (i) Show that the coordinates of A are (2, 4) 2
 - (ii) Hence find the area enclosed between the curves. 3
- (d) Using the trapezoidal rule with 4 subintervals, evaluate the area under the curve $y = x^x$ between $x = 1$ and $x = 3$, correct to 2 decimal places. 2

Question 14 (15 marks) Start a NEW page in your writing booklet.

- (a) A function is defined by $f(x) = \frac{x^3}{4}(x - 8)$
- (i) Find the coordinates of the stationary point(s) of the graph of $y = f(x)$ and determine their nature. 4
- (ii) Sketch the graph of $y = f(x)$ showing all its essential features including stationary points and intercepts. 2
- (iii) For what values of x is the curve increasing? 1

- (b) A rainwater tank with a volume of 9m^3 is installed in a new house. At 8am rain begins to fall and flows into the empty tank at the rate given by

$$\frac{dV}{dt} = \frac{36t}{t^2 + 20}$$

where t is the time in hours and V is the volume measured in cubic metres. ($t = 0$ is represented by 8am.)

- (i) Show by integration or otherwise that the volume of water in the tank at time, t is given by 2

$$V = 18 \log_e \left(\frac{t^2 + 20}{20} \right), t > 0$$

- (ii) Find the time when the tank will be completely filled with water (to the nearest minute). 3
- (iii) Later, when the tank is full and the rain has stopped, Louise turns on the pump which pumps the water out at the rate given by 3

$$\frac{dV}{dT} = \frac{T^2}{k}$$

where T is the time from when Louise turns on the pump and k is a constant. The pump continues for 5 hours until the tank is empty.

Find the value of k .

Question 15 (15 marks) Start a NEW page in your writing booklet.

- (a) A particle is moving along a straight line. Its displacement from a fixed point on the line at time t seconds is given by $x = 4t^3 - 3t^2 - 18t + 1$, where $t \geq 0$ and x is in metres.
- (i) Find the velocity v in terms of t . 1
 - (ii) Find the acceleration, a , in terms of t 1
 - (iii) At what time(s) does the particle come to rest? 1
 - (iv) Where does the particle come to rest? 1
 - (v) How far does the particle travel in the first 2 seconds? 2
- (b) Two cultures of bacteria are prepared in a laboratory. They are to be used to test the effectiveness of two drugs. One culture has 1000 bacteria and Drug A reduces this number to 250 in 5 minutes. The other culture has 1250 bacteria and Drug B reduces this number to 500 in 3 minutes. Both cultures are reduced according to the model $N = N_0 e^{-kt}$ where N is the number of bacteria and t is the time since the drug was administered in minutes.
- (i) Which drug is more effective in reducing the number of bacteria? Support your answer with calculations. 3
 - (ii) How long will it take for Drug B to reduce the second culture to 10% of its original number of bacteria? 2
- (c) (i) Show that $\frac{d}{dx} \left(x e^{\frac{x}{2}} \right) = e^{\frac{x}{2}} + \frac{1}{2} x e^{\frac{x}{2}}$ 2
- (ii) Hence find $\int x e^{\frac{x}{2}} dx$ 2

Question 16 (15 marks) Start a NEW page in your writing booklet.

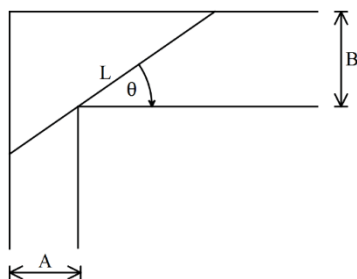
(a) Given $\tan A = \sqrt[3]{\frac{x}{y}}$ and $0 < A < \frac{\pi}{2}$

(i) Show that $\cos A = \frac{y^{\frac{1}{3}}}{\sqrt{x^{\frac{2}{3}} + y^{\frac{2}{3}}}}$ 2

(ii) Hence write down the value of $\sin A$ 1

- (b) Two corridors meet at right angles. One has a width of A metres, and the other has a width of B metres.

Mario the plumber wants to find the length of longest pipe, L that he can carry horizontally around the corner as seen in the diagram below. Assume that the pipe has negligible diameter.



(i) Show that $L = A \sec \theta + B \operatorname{cosec} \theta$ 1

- (ii) Explain why the length of the pipe, L , needs to be minimised in order to obtain the length of longest pipe that can be carried around the corner. 1

(iii) Hence show that when $\tan \theta = \sqrt[3]{\frac{B}{A}}$ the solution will be minimised. [You do **NOT** need to test to show that the solution will give a minimum length.] 4

(iv) Hence using the result in part (a), show that the length of the largest pipe that can be carried around the corner is $L = \left(A^{\frac{2}{3}} + B^{\frac{2}{3}}\right)^{\frac{3}{2}}$ 3

(c) The sequence 3

$$1, 2, 1, 2, 2, 1, 2, 2, 2, 1, 2, 2, 2, 2, 1, 2, 2, 2, 2, 2, 1, 2, \dots$$

consists of 1's separated by blocks of 2's with n 2's in the n th block.

Find the sum of the first 1234 terms in the sequence.

End of Examination

Multiple Choice

1. B 2. C 3. C 4. D 5. B 6. D 7. B 8. C 9. C 10. A

Question 11

$$a) \frac{1}{\sqrt{5}-\sqrt{2}} \times \frac{\sqrt{5}+\sqrt{2}}{\sqrt{5}+\sqrt{2}}$$

$$= \frac{\sqrt{5}+2}{5-2}$$

$$= \frac{\sqrt{5}+\sqrt{2}}{3}$$

$$b) |x-1| < 3$$

$$-3 < x-1 < 3$$

$$-2 < x < 4$$



$$c) 100^\circ = \frac{5\pi}{9} \text{ radians}$$

$$l = r\theta$$

$$= 12 \times \frac{5\pi}{9}$$

$$= \frac{20\pi}{3} \text{ cm}$$

$$d) (i) 2 \sec^2 2x$$

$$(ii) \frac{d}{dx} (x^2 \log_e x)$$

$$= x^2 \cdot \frac{1}{x} + \log_e x \cdot 2x$$

$$= x + 2x \log_e x$$

$$= x(1 + 2 \log_e x)$$

$$e) \int x^3 + \frac{1}{2x} dx$$

$$= \frac{x^4}{4} + \frac{1}{2} \log_e |x| + C$$

$$f) i) \int_0^{\frac{2\pi}{3}} \sin\left(\frac{x}{2}\right) dx$$

$$= \left[-2 \cos \frac{x}{2} \right]_0^{\frac{2\pi}{3}}$$

$$= -2 \left[\cos \frac{\pi}{3} - \cos 0 \right]$$

$$= -2 \left[\frac{1}{2} - 1 \right]$$

$$= 1$$

$$ii) \int_1^2 e^{4x} + e^{-x} dx$$

$$= \left[\frac{1}{4} e^{4x} - e^{-x} \right]_1^2$$

$$= \frac{1}{4} e^8 - \frac{1}{e^2} - \frac{1}{4} e^4 + \frac{1}{e}$$

Question 12.

a) i) In $\triangle CBF$ and $\triangle AEB$

$$\left. \begin{array}{l} \angle CBF = \angle BEA \text{ (alternate angles on parallel lines are equal, } AE \parallel BC) \\ \angle BCF = \angle EAB \text{ (opposite angles of parallelogram are equal)} \\ \angle ABF = \angle BFC \text{ (alternate angles on parallel lines are equal, } AB \parallel DC) \end{array} \right\} \begin{array}{l} \text{use} \\ \text{two} \end{array}$$

$\therefore \triangle CBF \equiv \triangle AEB$ (equiangular)

ii) $\frac{AB}{AE} = \frac{CF}{CB}$ (corresponding sides of similar triangles are in proportion)

$$\frac{AB}{15} = \frac{3}{7}$$

$$AB = 6\frac{3}{7} \text{ cm}$$

b) i) $m_{AC} = \frac{4}{3}$

$\therefore$ equation of AC: $y - 3 = \frac{4}{3}(x - 7)$

$$3y - 9 = 4x - 28$$

$$4x - 3y - 19 = 0$$

ii) $m_{BC} = \frac{2}{-1} = -2$

$$m_{AD} = \frac{9 - (-1)}{-1 - 4}$$

$$= \frac{10}{-5}$$

$$= -2$$

Since $m_{BC} = m_{AD}$ then $BC \parallel AD$

$$\text{iii) } m_{DC} = \frac{9-3}{-1-7}$$

$$= -\frac{3}{4}$$

$$\text{Since } m_{AC} \times m_{DC} = \frac{4}{3} \times -\frac{3}{4}$$

$$= -1$$

$$AC \perp DC$$

$$\text{iv) } AC = \sqrt{(7-4)^2 + (3-(-1))^2}$$

$$= \sqrt{9+16}$$

$$= 5$$

$$\text{v) } d = \left| \frac{4 \times 8 - 3 \times 1 - 19}{\sqrt{4^2 - (-3)^2}} \right|$$

$$= 2$$

$$\text{vi) } DC = \sqrt{8^2 + 6^2}$$

$$= 10$$

$$\text{Area of trapezium} = \text{Area of } \triangle ACD + \text{Area of } \triangle ACB$$

$$= \frac{1}{2} AC \times DC + \frac{1}{2} AC \times d$$

$$= \frac{5}{2} (10+2)$$

$$= 30 \text{ u}^2$$

Question 13

a) (i) $x^2 = 12y + 6x + 15$

$$x^2 - 6x = 12y + 15$$

$$\begin{aligned}(x-3)^2 &= 12y + 24 \\ &= 12(y+2)\end{aligned}$$

$\therefore$ vertex $(3, -2)$

(ii) Focal length: $a = 3$

$\therefore$ Focus: $(3, 1)$

(iii) $y = -5$

b) (i) Equal roots when $\Delta = 0$

$$\Delta = (-p)^2 - 4(p-1)$$

$$= p^2 - 4p + 4$$

$$= (p-2)^2$$

$$\Delta = 0 \Rightarrow p = 2$$

(ii) When $x = 3$ is a root

then $9 - 3p + p - 1 = 0$

$$8 - 2p = 0$$

$$2p = 8$$

$$p = 4$$

c) i) Solving the equations simultaneously

$$x^2 = 4x - x^2$$

$$2x^2 - 4x = 0$$

$$2x(x-2) = 0$$

$$\therefore x = 0 \text{ or } x = 2$$

$\therefore$ the x coordinate of A is 2.

$$\text{when } x = 2 \quad y = 2^2 = 4$$

$$\therefore A(2, 4)$$

$$(ii) \quad A = \int_0^2 (4x - x^2) - x^2 dx$$

$$= \int_0^2 4x - 2x^2 dx$$

$$= \left[2x^2 - \frac{2x^3}{3} \right]_0^2$$

$$= \left(2 \times 4 - \frac{2 \times 8}{3} \right)$$

$$= \frac{8}{3} \text{ u}^2$$

d)

x	1	1.5	2	2.5	3
y	1	1.84	4	9.88	27

$$A \doteq \frac{0.5}{2} \left[1 + 3^3 + 2(1.5^{1.5} + 2^2 + 2.5^{2.5}) \right]$$

$$\doteq 14.86 \text{ (to 2 decimal place)}$$

Question 14

$$\begin{aligned} \text{a) i) } f(x) &= \frac{x^3}{4}(x-8) \\ &= \frac{x^4}{4} - 2x^3 \end{aligned}$$

$$\begin{aligned} f'(x) &= x^3 - 6x^2 \\ &= x^2(x-6) \end{aligned}$$

Stationary points occur when $f'(x)=0$

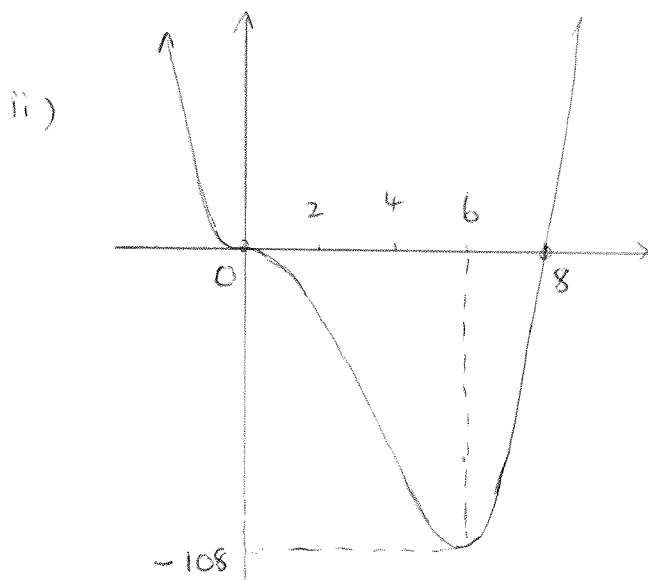
$$\therefore x^2(x-6)=0$$

$$\therefore x=0 \quad \text{OR} \quad x=6$$

x	-1	0	5	6	7
$f'(x)$	-7	0	-25	0	49
			/		/

$\therefore$ horizontal point of inflexion at $(0,0)$

and a absolute minimum turning point at $(6,-108)$



iii) Increasing when $x > 6$.

$$b) i) \frac{dV}{dt} = \frac{36t}{t^2+20}$$

$$V = \int \frac{36t}{t^2+20} dt$$

$$= 18 \int \frac{2t}{t^2+20} dt$$

$$= 18 \log_e(t^2+20) + C$$

$$\text{when } t=0 \quad V=0$$

$$\therefore C = -18 \log_e 20$$

$$\therefore V = 18 \log_e(t^2+20) - 18 \log_e 20$$

$$= 18 \log_e \left(\frac{t^2+20}{20} \right)$$

$$ii) \text{ Tank will be filled when } V=9$$

$$9 = 18 \log_e \left(\frac{t^2+20}{20} \right)$$

$$\frac{1}{2} = \log_e \left(\frac{t^2+20}{20} \right)$$

$$\frac{t^2+20}{20} = e^{\frac{1}{2}}$$

$$t^2 = 20e^{\frac{1}{2}} - 20$$

$$t = \sqrt{12.9744} \quad \text{as } t > 0$$

$$\doteq 3.6 \text{ hours or } 3 \text{ hrs and } 37 \text{ min.}$$

$$iii) V = \int \frac{T^2}{k} dT$$

$$V = \frac{T^3}{3k} + D$$

$$\text{when } T=0 \quad V=9 \quad \therefore D=9$$

$$V = \frac{T^3}{3k} + 9$$

if students

differentiate V to get $\frac{dV}{dt}$

they must test initial condition satisfies

$$\text{when } T=5 \quad V=0$$

$$\therefore 0 = \frac{5^3}{3k} + 9$$

$$\frac{-125}{3k} = -9$$

$$k = -\frac{125}{27}$$

Question 15

a) i) $v = 12t^2 - 6t - 18$

ii) $a = 24t - 6$

iii) $12t^2 - 6t - 18 = 0$

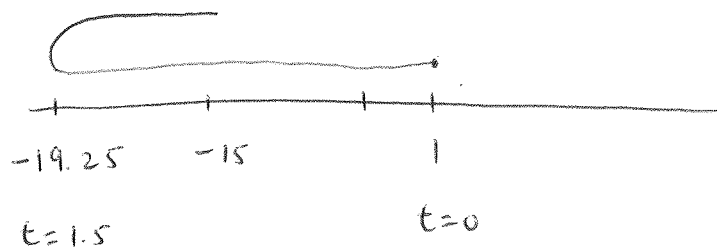
$$6(2t^2 - t - 3) = 0$$

$$6(2t - 3)(t + 1) = 0$$

$$\therefore t = 1.5 \quad \text{as } t \geq 0$$

iv) -19.25 (19.5m left of 0)

v)



when $t = 0$ $x = 1$

when $t = 1.5$ $x = -19.25$

when $t = 2$ $x = -15$

$\therefore$ total distance travelled:

$$20.25 + 4.22$$

$$= 24.5 \text{ m}$$

b) i) Drug A

$$250 = 1000 e^{-5k}$$

$$0.25 = e^{-5k}$$

$$-5k = \ln(0.25)$$

$$k = \frac{\ln(0.25)}{-5}$$

$$\hat{=} 0.277...$$

Drug B

$$500 = 1250 e^{-3k}$$

$$0.4 = e^{-3k}$$

$$-3k = \ln(0.4)$$

$$k = \frac{\ln(0.4)}{-3}$$

$$\hat{=} 0.305$$

The decay constant for drug B is larger than the one for drug A $\therefore$ drug B is more effective.

ii) $0.1 = e^{-0.305t}$

$$\ln(0.1) = -0.305t$$

$$t = \frac{\ln(0.1)}{-0.305}$$

$$\hat{=} 7.54 \text{ minutes.}$$

$$\begin{aligned} \text{c) i) } \frac{d}{dx} \left(x e^{\frac{x}{2}} \right) &= e^{\frac{x}{2}} \cdot 1 + x \cdot \frac{1}{2} \cdot e^{\frac{x}{2}} \\ &= e^{\frac{x}{2}} + \frac{x}{2} e^{\frac{x}{2}} \end{aligned}$$

$$u = x$$

$$u' = 1$$

$$v = e^{\frac{x}{2}}$$

$$v' = \frac{1}{2} e^{\frac{x}{2}}$$

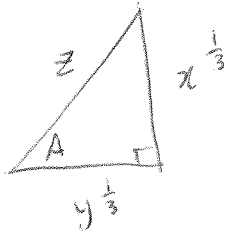
$$\text{ii) } x e^{\frac{x}{2}} = \int e^{\frac{x}{2}} dx + \int \frac{x}{2} e^{\frac{x}{2}} dx$$

$$\therefore \frac{1}{2} \int x e^{\frac{x}{2}} dx = x e^{\frac{x}{2}} - \int e^{\frac{x}{2}} dx$$

$$\begin{aligned} \int x e^{\frac{x}{2}} dx &= 2x e^{\frac{x}{2}} - 2 \cdot 2 e^{\frac{x}{2}} + C \\ &= 2(x-2) e^{\frac{x}{2}} + C \end{aligned}$$

Question 16

$$\begin{aligned} \text{a) i) } \tan A &= \frac{\sqrt[3]{x}}{\sqrt[3]{y}} \quad 0 < A < \frac{\pi}{2} \\ &= \frac{x^{\frac{1}{3}}}{y^{\frac{1}{3}}} \end{aligned}$$



let z be the hypotenuse

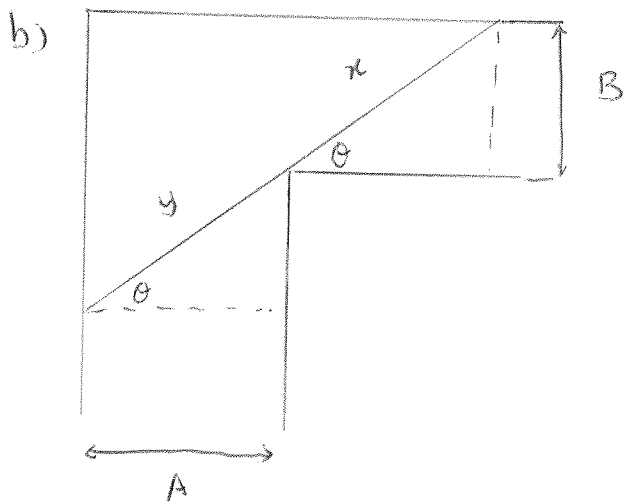
$$\left(x^{\frac{1}{3}}\right)^2 + \left(y^{\frac{1}{3}}\right)^2 = z^2 \quad [\text{Pythagoras' Thm}]$$

$$z = \sqrt{x^{\frac{2}{3}} + y^{\frac{2}{3}}}$$

$$\text{as } 0 < A < \frac{\pi}{2} \quad \cos A > 0$$

$$\cos A = \frac{y^{\frac{1}{3}}}{\sqrt{x^{\frac{2}{3}} + y^{\frac{2}{3}}}} \quad \text{where } y > 0.$$

$$\text{ii) } \sin A = \frac{x^{\frac{1}{3}}}{\sqrt{x^{\frac{2}{3}} + y^{\frac{2}{3}}}} \quad \text{where } x > 0$$



$$\text{Let } L = x + y$$

$$\sin \theta = \frac{B}{x}$$

$$x = B \operatorname{cosec} \theta$$

$$\cos \theta = \frac{A}{y}$$

$$\therefore y = A \sec \theta$$

$$\therefore L = B \operatorname{cosec} \theta + A \sec \theta$$

ii) If $\theta=0$ then pipe is completely in the corridor of width B and as $\theta \rightarrow 0$ then $L \rightarrow \infty$. Likewise if $\theta = \frac{\pi}{2}$ the pipe is completely in corridor with width A . So somewhere in the interval $0 < \theta < \frac{\pi}{2}$ is an angle that will minimise L . Any pipe larger than this L will not fit around the corner. Any pipe smaller than this will not be the largest pipe possible.

iii) $L = A \sec \theta + B \operatorname{cosec} \theta$

$$\frac{dL}{d\theta} = A \sec \theta \tan \theta - B \operatorname{cosec} \theta \cot \theta$$

$$= \frac{A \sin \theta}{\cos^2 \theta} - \frac{B \cos \theta}{\sin^2 \theta}$$

$$= \frac{A \sin^3 \theta - B \cos^3 \theta}{\sin^2 \theta \cos^2 \theta}$$

L is minimised when $\frac{dL}{d\theta} = 0$

i.e. when $A \sin^3 \theta - B \cos^3 \theta = 0$

$$A \sin^3 \theta = B \cos^3 \theta$$

$$\frac{\sin^3 \theta}{\cos^3 \theta} = \frac{B}{A}$$

$$\tan^3 \theta = \frac{B}{A}$$

$$\tan \theta = \sqrt[3]{\frac{B}{A}}$$

$\therefore \tan \theta = \sqrt[3]{\frac{B}{A}}$ will give the minimum solution.

iv) Since $L = A \sec \theta + B \operatorname{cosec} \theta$

$$= \frac{A}{\cos \theta} + \frac{B}{\sin \theta}$$

when $\tan \theta = \sqrt[3]{\frac{B}{A}}$

$$\cos \theta = \frac{A^{\frac{1}{3}}}{\sqrt{A^{\frac{2}{3}} + B^{\frac{2}{3}}}}$$

from part a (i)

and $\sin \theta = \frac{B^{\frac{1}{3}}}{\sqrt{A^{\frac{2}{3}} + B^{\frac{2}{3}}}}$

from part a (ii)

then
$$L = \frac{A \sqrt{A^{\frac{2}{3}} + B^{\frac{2}{3}}}}{A^{\frac{1}{3}}} + \frac{B \sqrt{A^{\frac{2}{3}} + B^{\frac{2}{3}}}}{B^{\frac{1}{3}}}$$

$$= A^{\frac{2}{3}} (A^{\frac{2}{3}} + B^{\frac{2}{3}})^{\frac{1}{2}} + B^{\frac{2}{3}} (A^{\frac{2}{3}} + B^{\frac{2}{3}})^{\frac{1}{2}}$$

$$= (A^{\frac{2}{3}} + B^{\frac{2}{3}}) (A^{\frac{2}{3}} + B^{\frac{2}{3}})^{\frac{1}{2}} \quad \left[\text{factorising out } (A^{\frac{2}{3}} + B^{\frac{2}{3}}) \right]$$

$$= (A^{\frac{2}{3}} + B^{\frac{2}{3}})^{\frac{3}{2}}$$

c) Consider $(1, 2)$ $(1, 2, 2)$ $(1, 2, 2, 2) \dots$
2 terms 3 terms 4 terms

Suppose the $(n-1)^{\text{th}}$ group is $(1, \underbrace{2, 2, 2, \dots, 2}_{n-1 \text{ terms}})$

Then $2 + 3 + 4 + \dots + n \leq 1234$

$$\therefore \frac{n-1}{2} [4 + (n-2)] \leq 1234$$

$$n^2 + n - 2 \leq 2468$$

$$n(n+1) \leq 2470$$

we can see that when $n=49$ 49×50 is close to 2470.

$\therefore n=49$ goes up to the 48^{th} group.

$$1234 - \frac{48 \times 49}{2} = 10$$

$\therefore$ the 49^{th} group can only have 1 1's and 9 2's

There are 49 1's in the first 1234 terms

and $(1 + 2 + 3 + \dots + 48) + 9$ 2's in the first 1234 terms.

$$= \left(\frac{48}{2} [2 + 47] + 9 \right) \quad 2's$$

$$= 1185 \quad 2's.$$

$\therefore$ sum of first 1234 terms is

$$49 + (1185 \times 2)$$

$$= 2419.$$